

WellnessBot

Installation & User Guide

Version: v2.0

Application: WellnessBot — Rehabilitation Decision Support

Document Type: Appendix — Installation & User Guide

Contents

1. System Overview	3
2. Prerequisites	4
3. Installation	5
3.1 Extract the Zip Archive	5
3.2 Create a Virtual Environment	6
3.3 Install Dependencies	6
3.4 Configure Environment Variables	7
4. Running the Application.....	8
4.1 Windows (PowerShell)	8
4.2 macOS / Linux	8
Accessing the Application	8
5. User Guide	9
5.1 Launching the App	9
5.2 Creating a User Profile	9
5.3 Starting a Conversation.....	10
5.4 Slot-Filling Dialogue Flow.....	11
5.5 System Responses.....	12
5.6 Providing Feedback.....	13
5.7 Knowledge Loop Analysis	14
6. Environment Variables Reference.....	15
7. Troubleshooting	16
7.1 Script not found / CommandNotFoundException	16
7.2 PowerShell script execution blocked	16
7.3 Application does not start.....	16
7.4 ModuleNotFoundError: No module named 'SystemCode'	16
7.5 openai.AuthenticationError or 401 Unauthorized	16
7.6 Profile not saving.....	16
7.7 Logs not being written	16

1. System Overview

WellnessBot is a neuro-symbolic hybrid AI chatbot that provides patients with contextually appropriate, evidence-grounded exercise recommendations during arthroscopic knee rehabilitation. It adapts dynamically to each user's surgery type, recovery phase, reported symptoms, available equipment, and exercise history.

Pipeline Architecture

User Input

- [1] NLU Extraction (OpenAI GPT-4o-mini)
- [2] Dialog Merge (Slot filling into ConversationState)
- [3] CLARIFY Check (Missing slots trigger question loop)
- [4] User State Inference (Derives phase_id, risk_flags)
- [5] Rule Engine (RECOMMEND / CLARIFY / FORBID / ESCALATE)
- [6] Intervention Planner (Exercise selection — RECOMMEND path only)
- [7] RAG Retrieval (BM25 evidence chunk lookup)
- [8] LLM Response Generation (Patient-facing validated output)
- User Output (Streamlit chat UI)

2. Prerequisites

Before installing WellnessBot, ensure the following are available on your machine:

Requirement	Minimum Version	Notes
Python	3.10+	Check with <code>python --version</code>
pip	Latest	Bundled with Python
OpenAI API Key	—	OpenAI API Key is required for full system functionality
Internet connection	—	Internet connection is required for OpenAI API calls

Note: WellnessBot has been tested on Windows 11. It is compatible with macOS and Linux, with minor command differences noted where applicable.

3. Installation

3.1 Extract the Zip Archive

The submission is delivered as a zip file. Extract it to a location of your choice.

After extraction, the folder layout will look like this:

IRS-PM-2026-01-24-GRP7-WellnessBot/

```
|— ProjectReport/
|   |— Installation_and_User_Guide ← this document
|   |— SystemCode/ ← all runnable code lives here
|       |— .env.example
|       |— requirements.txt
|       |— run_streamlit.ps1
|       |— run_tests.ps1
|       |— data/
|       |— logs/
|       |— wellnessbot/
```

Important: All terminal commands in this guide must be run from inside the SystemCode/ folder unless otherwise stated.

Open a terminal (PowerShell on Windows, or Terminal on macOS/Linux) and navigate there:

Windows (PowerShell):

```
cd path\to\IRS-PM-2026-01-24-GRP7-WellnessBot\SystemCode
```

macOS / Linux:

```
cd path/to/IRS-PM-2026-01-24-GRP7-WellnessBot/SystemCode
```

3.2 Create a Virtual Environment

It is strongly recommended to use a virtual environment to isolate dependencies. Run the following from inside SystemCode/:

Windows:

```
python -m venv .venv
.venv\Scripts\activate
```

macOS / Linux:

```
python -m venv .venv
source .venv/bin/activate
```

You should see (.venv) prepended to your terminal prompt once the environment is active.

3.3 Install Dependencies

With the virtual environment active, install all required packages:

```
pip install -r requirements.txt
```

This installs the following packages:

Package	Version	Purpose
streamlit	$\geq 1.36.0$	Web UI framework
openai	$\geq 1.40.0$	LLM API client (GPT-4o-mini)
pydantic	$\geq 2.7.0$	Data validation and schema enforcement
python-dotenv	$\geq 1.0.1$	Environment variable loading from .env
pyyaml	$\geq 6.0.1$	YAML configuration parsing
pytest	$\geq 8.2.0$	Test suite runner
ruff	$\geq 0.6.0$	Code linter

3.4 Configure Environment Variables

The .env file must be created inside SystemCode/ (alongside requirements.txt). Create it by copying the provided example:

Windows (PowerShell):

```
Copy-Item .env.example .env
```

macOS / Linux:

```
cp .env.example .env
```

Open SystemCode/.env in a text editor and set the required values:

```
APP_VERSION=v2.0
LOGGING_ENABLED=1
LOG_DIR=./logs
OPENAI_API_KEY=sk-proj-xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
OPENAI_MODEL=gpt-4o-mini
```

Security: Please do not share your OPENAI Key

4. Running the Application

Note: All commands below assume your terminal is inside `SystemCode/` with the virtual environment active.

4.1 Windows (PowerShell)

A convenience script is provided that automatically sets `PYTHONPATH` and launches Streamlit. Run it from inside `SystemCode/`:

```
.\run_streamlit.ps1
```

4.2 macOS / Linux

From inside `SystemCode/`, with the virtual environment active:

```
PYTHONPATH=.. streamlit run wellnessbot/Wellnessbot.py
```

Accessing the Application

Once started, open your web browser and navigate to:

```
http://localhost:8501
```

Streamlit will also print the URL to the terminal. The application loads immediately — no build step is required.

5. User Guide

5.1 Launching the App

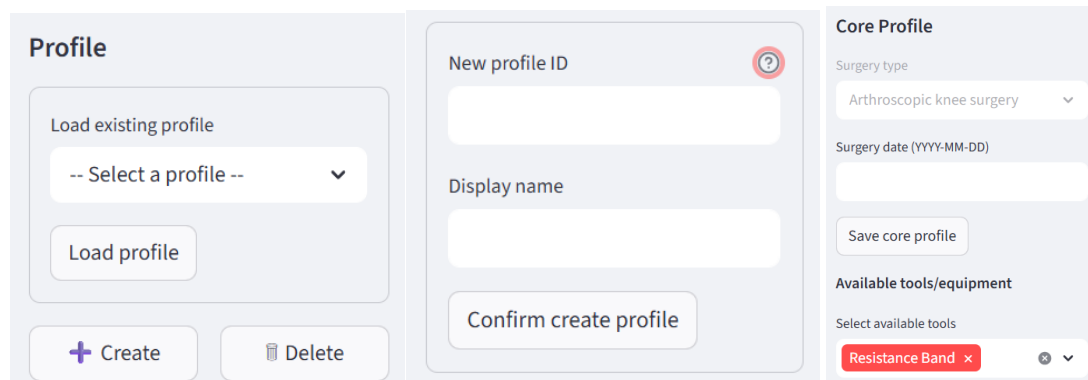
After running the start command (see section 4), open <http://localhost:8501> in your browser. You will see the WellnessBot chat interface with a sidebar on the left.

5.2 Creating a User Profile

Before starting a conversation, you must select or create a user profile. The profile stores your surgery details and equipment list, so they persist across sessions.

Steps:

1. In the left sidebar, locate the Profile section.
2. If no profiles exist, click Create.
3. Enter the Display Name, then click “Confirm create profile” to initialise the profile.
4. After the profile is created, input the Surgery Date and Equipment Available in the profile settings.
5. To switch profiles, select the desired profile from the dropdown menu in the sidebar.



The screenshot displays the 'Profile' management interface, which is divided into three main sections:

- Profile (Left):** Contains a 'Load existing profile' section with a dropdown menu labeled '-- Select a profile --' and a 'Load profile' button. At the bottom are '+ Create' and 'Delete' buttons.
- New profile ID (Middle):** A form for creating a new profile, featuring a 'New profile ID' field with a red question mark icon, a 'Display name' field, and a 'Confirm create profile' button.
- Core Profile (Right):** A form for editing profile details, including a 'Surgery type' dropdown (currently set to 'Arthroscopic knee surgery'), a 'Surgery date (YYYY-MM-DD)' field, a 'Save core profile' button, and an 'Available tools/equipment' section with a 'Select available tools' dropdown (currently showing 'Resistance Band' with a red 'x' icon).

Profile data is stored locally in `SystemCode/data/profiles/<profile_id>.json`. It is not transmitted anywhere other than the OpenAI API for NLU processing.

5.3 Starting a Conversation

With a profile selected, type your message in the chat input box at the bottom of the screen and press Enter or click Send.

Example opening messages:

- “None”
- “I am having slight fever and no bleeding”

Rehabilitation Decision Support

Evidence-based rehab recommendations tailored to your recovery stage. Not a medical advice.

End conversation / RestartClear chat only

Welcome to the Wellnessbot, Patient_1A.

I can help suggest suitable rehabilitation exercises based on your recovery stage.

Please select your available tools/equipment if you have not done so.

Current phase: P1_2 (Early)

Are you having any symptoms today, such as fever or excessive bleeding? If none, just say 'none'.

> Developer details



The bot will respond based on the information it has collected so far. If information is missing, it will ask follow-up questions.

5.4 Slot-Filling Dialogue Flow

WellnessBot collects the required user information before generating any exercise recommendation. If any required field is missing, the system issues a targeted follow-up question to ensure completeness of the input state.

For example, if the surgery date has not been provided during profile setup, WellnessBot will prompt the user to enter the surgery date before proceeding to the symptom assessment stage.

Slot	What the Bot Asks	Example Answer
Surgery date / weeks	"When was your surgery or injury? Please tell me the date (YYYY-MM-DD) "	"2025-12-01"
Symptoms	"Are you having any symptoms today, such as fever or excessive bleeding? If none, just say “none”."	"I am having fever" / "None"
Pain score	"How would you rate the pain from 0 to 3?"	Numeric input (e.g., 1) or selection via UI buttons
Swelling score	"How would you rate the swelling from 0 to 3? "	Numeric input (e.g., 0) or selection via UI buttons

Tips:

- If you have already set your surgery type and date in your profile, the bot may skip those questions.
- The bot remembers your answers within a session and will not re-ask completed slots.

5.5 System Responses

Once sufficient information is collected, WellnessBot generates one of four responses:

RECOMMEND — Exercise Recommendation

The most common response. The bot provides:

- Self Care (if applicable) — ice/elevation advice based on your symptoms
- Recommended Exercise — exercise name, position, equipment needed, and caution notes
- How To Perform — one grounded instruction line from the clinical evidence database
- References — source citations with links

 **SELF CARE**

Supportive self-care guidance: ice and elevate for 20 minutes (3 to 4 times per day).

RECOMMENDED EXERCISE

Hip Abduction

HOW TO PERFORM

Lie on your unoperated side with your knees fully extended. Raise the operated limb upward to a 45-degree angle as illustrated. Hold for one second, then lower it slowly. Repeat 20 times.

REFERENCES

(S4: https://sportmed.com/wp-content/uploads/Knee_Scope_21.pdf)

CLARIFY — Follow-up Question (Internal Step)

The bot asks for one missing piece of information before it can make a recommendation. A list of possible exercises may also be shown.

 How would you rate the pain from 0 to 3? (0 = none, 1 = mild, 2 = moderate, 3 = severe)

> Developer details

Select pain level:

None

Mild (1)

Moderate (2)

Severe (3)

FORBID — Exercise Blocked

The bot explains why an exercise cannot be recommended at this time (e.g., your swelling score is too high). No exercise is provided.

 Do not proceed with exercise yet. Supportive follow-up required first: use pain relief as prescribed, stop exercise for now.

Supportive care: Supportive self-care guidance: ice and elevate for 20 minutes (3 4 times per day)

ESCALATE — Seek Clinical Help

If red-flag symptoms are detected (e.g., fever, active bleeding, extreme swelling), the bot advises you to contact your healthcare provider immediately. No exercise is recommended.

 Exercise is not advised at this time. It is important to speak with a clinician before beginning any physical routine.

5.6 Providing Feedback

After each final system response (RECOMMEND, FORBID, or ESCALATE), a feedback prompt is presented to the user in the form of a thumbs-up or thumbs-down selection. Users may indicate whether the response was helpful or not through this interface.

All feedback is recorded as part of the system's audit trail, together with the corresponding decision, input state, and reasoning outputs. This information is subsequently used for developer analysis to assess system performance, identify failure patterns, and support targeted refinement of the knowledge graph, rule set, and planner logic through the offline knowledge loop.

 **Chat ended.**

Please click **End conversation / Restart** or **Clear chat only** to start a new case.

5.7 Knowledge Loop Analysis

The Knowledge Loop Analysis page is used to evaluate system behaviour using interaction logs and user feedback.

The screenshot displays the 'Knowledge Loop Analysis' interface. On the left is a sidebar with 'Wellnessbot' and 'Knowledge Loop Analysis' tabs. The main area is titled 'Input files' and shows paths for 'Interactions JSONL path' and 'Feedback JSONL path', both pointing to files in 'C:\Users\Collins\Documents\VS Workspace\WellnessBot_v2\SystemCode\logs'. A 'Run mining' button is present. Below this, a green bar indicates 'Mining complete.'.

The 'Basic summary' section shows four metrics: Interactions (2), Matched feedback (1), Unmatched feedback (7), and Candidate cases (2).

The 'Candidate cases' section contains a table with columns: case_type, priority, title, description, evidence_count, sample_interaction_ids, and proposal_review. It lists two cases related to rule combinations and planner selection.

The 'KG rule mining' section shows a table with columns: rule_combination, count, thumbs_down, thumbs_down_rate, top_expected_actions, sample_comments, and sample_interaction_ids. It lists two rule combinations.

The 'Safety consistency' section shows a table with columns: red_flag_term, phase_id, action_distribution, and sample_interaction_ids. It lists 'excessive bleeding' and 'PL_1'.

The 'Planner mining' section shows a table with columns: exercise_id, exercise_name, phase_id, priority, count, thumbs_down, thumbs_down_rate, top_expected_actions, sample_comments, and sample_interaction_ids. It lists 'PL_ES' and 'Single leg squats'.

Steps:

1. Select Input Files
Ensure the interaction logs and feedback logs are correctly loaded.
2. Run Mining
Click “Run Mining” to process logs and generate results.
3. Review Basic Summary
View key metrics: total interactions, feedback coverage, and refinement candidates.
4. Analyse Refinement Candidates
Review flagged cases with issue summaries, interaction IDs, and suggested actions.
These highlight potential problems such as rule conflicts, negative feedback patterns, or suboptimal planner decisions.
5. Deep Dive into Mining Views
 - KG Rule Mining: Identify problematic rule combinations
 - Safety Consistency: Check if red-flag conditions trigger correct actions
 - Planner Mining: Evaluate exercise selection behaviour

This workflow supports systematic identification of system weaknesses and guides manual, developer-driven refinement of the knowledge graph, rules, and planner, without affecting runtime logic.

6. Environment Variables Reference

Variable	Required	Default	Description
OPENAI_API_KEY	Yes	—	OpenAI API key for NLU and LLM response generation
OPENAI_MODEL	No	gpt-4o-mini	OpenAI model for NLU extraction
LOGGING_ENABLED	No	1	Set to 0 to disable interaction logging
LOG_DIR	No	./logs	Directory for daily JSONL interaction log files
APP_VERSION	No	v2	Version tag written into audit log entries

7. Troubleshooting

7.1 Script not found / CommandNotFoundException

Ensure you are running the command from inside SystemCode/, not from the repo root or elsewhere:

```
cd path\to\IRS-PM-2026-01-24-GRP7-WellnessBot\SystemCode .\run_streamlit.ps1
```

7.2 PowerShell script execution blocked

Run this once to allow local scripts, then retry:

```
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
```

7.3 Application does not start

- Ensure your virtual environment is active ((.venv) in your terminal prompt).
- Ensure all dependencies are installed: `pip install -r requirements.txt`.
- Ensure you are running the command from inside SystemCode/.

7.4 ModuleNotFoundError: No module named 'SystemCode'

- PYTHONPATH must point to the parent of SystemCode/ (i.e., IRS-PM-2026-01-24-GRP7-WellnessBot/).
- On macOS/Linux, use PYTHONPATH=.. when running commands manually.
- On Windows, run_streamlit.ps1 sets this automatically — use the script rather than running streamlit directly.

7.5 openai.AuthenticationError or 401 Unauthorized

- Your OPENAI_API_KEY in SystemCode/.env is missing or incorrect.
- Verify the key at platform.openai.com.

7.6 Profile not saving

- Ensure the SystemCode/data/profiles/ directory exists. It is created automatically on first run.
- Check write permissions on the SystemCode/data/ directory.

7.7 Logs not being written

- Ensure LOGGING_ENABLED=1 in SystemCode/.env.
- Ensure the SystemCode/logs/ directory exists and is writable.